

as the voltage on C1 is greater than FET gate's threshold voltage. The time period of the alarm depends on the value of capacitor C1 and resistor R2, which governs the capacitor's discharge current.

The component values provided here give an on-time of about ten minutes. You can increase or decrease the time period of alarm by changing the values of capacitor C1 or resistor R1.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Press switch S2 along with your electrical appliance's switch to enable the alert alarm. Switch S1 is power-on switch. ●

OVER-TEMPERATURE ALARM

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This circuit can be used as a warning device which triggers an alarm whenever the temperature exceeds a preset level. It can also be used as a door-alert or general warning signal.

The collector of transistor T2 goes high when thermistor's temperature becomes higher than preset temperature level. High voltage of collector of T2 is used to enable the pulse oscillator.

Pulse oscillator section. This section is built around gates N1 and N2. The NAND gate N1 has one of its

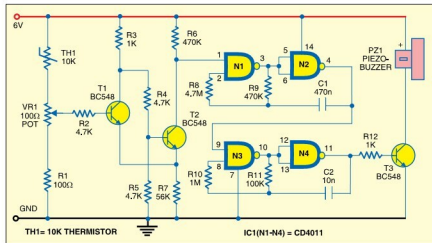


Audio oscillator section. This section is built around gates N3 and N4. The output of the pulse oscillator is fed to input pin 9 of gate N3. When output of pulse oscillator goes high, audio oscillator starts oscillating. This frequency is given to the base of transistor T3 to drive piezobuzzer PZ1, which starts producing an intermittent beep sound. The beep rate depends on the values of capacitor C2 and resistor R11.

At normal condition, thermistor TH1 keeps the base of transistor T1 low. Transistor T1 is cut off and T2 conducts, keeping collector voltage of T1 high while collector voltage of T2 is low. As a result both the oscillators are disabled and piezobuzzer remains silent.

On the other hand, when temperature of thermistor goes high, the base of transistor T1 also goes high. Transistor T1 conducts and T2 cuts off, resulting in collector voltages of T1 as low and T2 high. Both oscillators oscillate and piezobuzzer produces intermittent sounds.

Assemble the circuit on a general-purpose PCB and enclose it in a suitable cabinet. Connect the thermistor TH1 using two wires and preset on the front panel. Mount the piezobuzzer firmly on the enclosure. Use 6V battery or a 6V adaptor to operate the circuit. Alternatively, you can use a 9V PP3-type battery also. ●



The circuit can be divided into three sections:

Control section. This section enables/disables the pulse oscillator depending on the temperature produced near the thermistor, which is used as a sensor. Thermistor used here is a heat-sensitive device that acts as a variable resistor. Around normal room temperature, its resistance remains high, but decreases with increase in temperature. The wiper of potentiometer VR1 is connected to the base of transistor T1.

inputs (pin 1) used for control. It oscillates when controlling section provides high voltage at pin 1 of gate N1. On the other hand it does not oscillate when control input pin 1 of gate N1 is held at logic low. The timing capacitor and resistor values of the pulse oscillator can be calculated as follows:

$$f = 1/(2.2 \times R9 \times C1)$$

A very low-frequency (pulsed oscillator) output is available at pin 4 of gate N2, used to enable/disable the audio oscillator.